Appendix A. CREZ and OSW Locations

Summarized in the table below are the specific CREZ and OSW coordinates. CREZ coordinates are taken at the centroid while OSW coordinates are taken at the point of most potential.

Table A1
CREZ Centroid Coordinates Per Grid

CREZ	Centroid Latitude	Centroid Longitude
	Luzon	
L1	15.3932	120.7703
L2	15.0321	120.6782
L3	15.3911	120.2106
L4	15.6640	120.7696
L5	16.2025	120.0192
L6	18.2852	120.8563
L7	18.1048	121.7055
L8	17.5620	121.7050
L9	16.7485	121.6106
L10	13.7691	122.4435
L11	14.0396	122.4445
L12	13.8576	122.8148
L13	12.9499	123.7340
	Visayas	
N1	10.8809	122.9828
N2	10.6107	122.8897
Py1	11.4231	122.8023
Py2	10.9721	122.8000
S1	12.2135	125.1096
B1	9.9682	124.2561
	Mindanao	
M2	7.8500	122.8500
M4	7.2000	126.3000
M7	8.1500	125.0800
M8	6.2500	125.1700

Table A2
OSW Sites Coordinates

OSW Site	Latitude	Longitude
	Luzon	
Northern Luzon	18.6427	120.9443
Manila Bay	14.4865	120.2433
	Visayas	
Guimaras Strait	10.7501	122.7798
Negros-Panay	10.3922	121.9984

Appendix B. Detailed Equations

This section provides detailed mathematical formulations, necessary equations, and corresponding nomenclature used to conduct the simulations.

Table B1
Descriptions of the sets, variables and parameters

Sets	Description
R	CREZ Locations (L1, L2, L3…)
P	Grid (Luzon, Visayas, Mindanao)
Y	Weather-Year (1985-2024)
I	VRE (Solar, Wind, Offshore Wind)
H	Hours (1-24)
PDP	Target Years (2025, 2030, 2040, 2050)
Variables	
$z_{r,i}$	installed-capacity potential of zone *r* for source *i*
$g_{r,h,i}$	hourly capacity factor of zone *r* for source *i*
$d_{pdp,h}$	scaled hourly projection for a single target year
$c_{pdp,i}$	corresponding installed capacity targets from PDP 2023-2050

Table B2
Equations used

Equation Number	**Equation**	Description
1	$w_{r,i} = \frac{z_{r,i}}{\sum_{r \in R} z_{r,i}}$	weight of CREZ *r* for source *i* (solar, wind, offshore)
2	$g_{composite,p,h,i} = \sum_{r \in R} (g_{r,h,i} \times w_{r,i})$	hourly composite capacity factor for source *i* in grid *p*
3	$n_{y,h} = d_{pdp,h} - \sum_{i \in I} (g_{composite,h,i} \times c_{pdp,i})$	net load for weather-year *y*